

Abhik ROYCHOUDHURY

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LinkedIn

Provost's Chair Professor
National University of Singapore
&
Editor-in-Chief
ACM TOSEM.

Research Interests: Trusted Agentic AI, AI for Code, Program Analysis, Software Security.

Education Ph.D. in Computer Science, State University of New York at Stony Brook, 2000.
Thesis: *Program Transformations for Verifying Parameterized Systems*

Employment National University of Singapore (NUS), School of Computing, since 2001.

- Professor, Computer Science Department (2014 - now).
  - Provost's Chair Professor (2020 – now).
  - Senior Scientific Advisor, Sonar, 2025-2026 after AutoCodeRover acquisition.
- Associate Professor, Computer Science Department (2007-2014).
- Assistant Professor, Computer Science Department (2001-2007).

Recent Projects and Initiatives

- "AI for Program Reasoning", NRF AI for Science, Co-Lead PI, ~$10M, 2026-30.
- "Agentic AI based Software of the future", Investigator Award (sole PI), ~$3M, 2026-31.
- "Fuzz Testing", CSA National Cybersecurity R&D program, Lead PI, $6.7M, 2023-27.
- "Automated Program Repair", Ministry of Education Tier 3, Lead PI, $7.5M, 2022-27.

Selected Publications (around 5)

- SemFix: Program Repair via Semantic Analysis, ICSE 2013.
- Coverage-based Greybox Fuzzing as Markov Chain, CCS 2016.
- Directed Greybox Fuzzing, CCS 2017.
- Automated Program Repair, Communications of the ACM, 2019.
- AutoCodeRover: Autonomous Program Improvement, ISSTA 2024.
- SpecRover: Code Intent Extraction via LLMs, ICSE 2025.

Recent Honors and Awards

- NUS University Research Recognition Award, 2026, given to 1-2 individuals annually, across all disciplines.
- Inaugural class of ACM SIGSOFT Software Engineering Academy 2026, for lasting contributions to 50 years of software engineering, in research, practice and education.
- Fellow of the ACM, 2024, "*contributions to automated program repair and fuzz testing.*"
- NUS Outstanding Graduate Mentor Award (inaugural recipient), 2024.
- 2023 ICSE Most Influential Paper Award (Test-of-Time) for ICSE 2013 paper SemFix.
- IEEE TCSE New Directions Award 2022 (awarded jointly with Cristian Cadar).

Significant Professional Appointments

- Editor-in-Chief, ACM Trans. on Software Engg. and Methodology (TOSEM) 2025-27.
- Steering Comm. Chair, ACM International Conference on Foundations of Software Engineering (FSE), from 2023. Served as General Chair, FSE 2022, held in Singapore.
- Program Co-chair, International Conference on Software Engineering (ICSE) 2024.
- Program Chair, Intl. Symposium on Software Testing and Analysis (ISSTA) 2016.
- Associate Editor, Communications of the ACM, from 2023.
- Chaired many award committees including ACM SIGSOFT Impact Paper Award 2024-25 and ACM SIGSAC Doctoral Dissertation Award in Security and Privacy, 2025.

Translational Impact

- AutoCodeRover, NUS spin-off on AI Software Engineer acquired by SonarSource, leading to a commercial product SonarQube Remediation Agent (announced 2026).
- AFLFast and AFLGo as extended grey-box fuzzing tools, for detecting program vulnerabilities. AFLFast has been integrated into the AFL++ distribution. The technology has been taken forward for agent assurance via Verigrey spinoff.

Sample PhD Student Placement

- Ruijie Meng, PhD NUS → Univ. of Waterloo (joining), Assistant Professor, IMDA Prize for Best PhD Thesis at NUS Computing.
- Ridwan Shariffdeen, PhD NUS → Principal Research Scientist, SonarSource Singapore.
- Van Thuan Pham, PhD NUS → University of Melbourne, Australia, Senior Lecturer.
- Marcel Böhme, PhD NUS → Max-Planck Institute (MPI), Germany, faculty member.
- Xiang Gao, PhD NUS → Beihang University, China, as Associate Professor.
- Shin Hwei Tan, PhD NUS → Concordia Univ, Canada, Chaired Associate Professor, Winner of ACM-W Rising Star Award, given to one early career female faculty member.
- Dawei Qi, PhD NUS → CTO, Shenzhen Secidea Network Security Tech. Ltd., China.
- Zhiyu Fan, PhD NUS → Senior Researcher, Microsoft Research Cambridge, UK.

Recent Invited Lectures and Keynotes

- Keynote speech, “Agentic AI for Software”, ACM India Annual Event, Hyderabad, Feb26.
- Plenary Speech, “Agentic AI for Software: Lessons in Trust”, InnovFest Suzhou, Oct 25.
- Keynote, Software Engineering for Machine Learning Applications (SEMLA) 2025.
- Keynote, 16th International Conference on Internetware, Trondheim, Norway, 2025.
- Recent Distinguished Lectures at Max Planck Inst. (2024), TU Delft Netherlands (2025).
- Invited Tutorial, Embedded Systems Week, October 2022.
- *Past Keynotes include*: 17th Innovations in Software Engineering Conference (ISEC) 24, 27th Asia-Pacific Software Engg. Conf. (APSEC) 20.

Teaching Authored a textbook “*Embedded Systems and Software Validation*” under Elsevier in 2009. Introduced a class “AI-driven innovation from Deep tech research” at NUS involving conversations with tech innovators, business co-founders and venture capitalists at NUS.

Citations *https://scholar.google.com.sg/citations?user=UxFWSJIAAAAJ&hl=en*